ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** magnet, add water and filter, evaporate filtrate	Each step removes one component.
2	**(a)** 30 g	Same as initial.
3	**(a)** it makes tiny particles clump and settle	Coagulation.
4	**(c)** sand	Insoluble.
5	**(b)** 18 g	Half.
6	**(c)** filtration cannot remove all germs	Disinfection is needed.
7	**(d)** fine clay	Smaller particles settle slowest.
8	**(d)** 7 g	25 - 18.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The difference in weight is used with wind, so R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they are different mixtures and methods; R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Some salt stuck to the filter or vessel (1); Handling losses (1).	1+1
12	Dissolve the sugar in water and strain out the grains (1); Evaporate the water to recover sugar (1).	1+1
13	Strain through cloth (1); Warm gently (1).	1+1
14	Sieve holes are large (1); Filter paper holes are tiny (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Magnet for iron (1); Water: wooden chips float, skim (1); Filter sand and chalk; evaporate salt (1).	1+1+1
16	Immiscible (1); Kerosene lighter (1); Floats on top (1).	1+1+1
17	Holds solid (1); Whey passes (1); Straining (1).	1+1+1
	OR: 36 g in 100 g (1); 90/36 x 100 (1); 250 g water (1).	1+1+1
18	Separates whey (1); Solid curd stays (1); Liquid passes (1).	1+1+1
	OR: Add water (1); Oil floats (1); Filter sand (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Threshing (1). (ii) Winnowing (1). (iii) Sieving (or handpicking) (1). (iv) Winnowing needs a dry mixture of light and heavy solids, and wind cannot separate a solid from a liquid (1); sand in water is separated by sedimentation and filtration (1).	1+1+1+2
20	Let sand settle, decant (1); Filter remaining (1); Dry the sand (1); Evaporate part of the filtrate for salt (1); Condense vapour for water (1).	1+1+1+1+1
	OR: Winnow to remove husk (1); Magnet for iron (1); Handpick stones (1); Wash if needed (1); Clean rice (1).	1+1+1+1+1

Mistakes to avoid:

- Choosing filtration for salt in water; dissolved salt passes through the filter, so **evaporation** is needed.
- Mixing **sedimentation** (settling) with **decantation** (pouring off the clear liquid).
- Mixing **threshing** (beating stalks to free grain) with winnowing (blowing away husk).
- Calling the liquid that passes through filter paper the residue; it is the **filtrate**.
- Using a magnet to separate sand or salt; only magnetic substances like iron are attracted.
- Believing that filtered or settled water is always safe to drink.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 20 g	Half of 40.
2	**(c)** to avoid disturbing the sediment	Keeps the sediment.
3	**(d)** they mix completely	One layer only.
4	**(a)** both particles are fine	Sizes are similar.
5	**(b)** funnel	Directing.
6	**(c)** churning	Churning makes lighter cream separate from milk.
7	**(b)** sieving	Different sizes.
8	**(d)** they scatter light strongly like muddy water	True solutions do not scatter much.
9	**(d)** A is false but R is true.	A is false: filtration cannot separate a dissolved solid. R is true, which is why evaporation is used.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Small flour particles pass through the holes while larger bran stays on the sieve.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cream is lighter than the rest of the milk (1); It forms a layer on top (1).	1+1
12	Dissolve, filter and evaporate (1); Crystallise (1).	1+1
13	Gold is heavy and settles (1); Lighter sand is washed away (1).	1+1
14	Let it stand so the oil floats, then pour it off (1); Oil is lighter than water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Powder dissolves faster (1); Stirring helps (1); Warm water dissolves faster (1).	1+1+1
16	Heat a drop on glass (1); Pure water leaves nothing (1); Salt water leaves a white solid (1).	1+1+1
17	Heat the solution in a plate with a cool lid above (1); Droplets of water collect on the lid (1); White salt is left in the plate (1).	1+1+1
	OR: Same water temperature and amount (1); Add spoon by spoon (1); Count spoons till saturated (1).	1+1+1
18	Settle and decant (1); Cloth or sand filter (1); Boil (1).	1+1+1
	OR: Filter bag traps dust (1); Air passes (1); Similar to filtration (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sedimentation (1). (ii) Decantation (1). (iii) Filtration (1). (iv) It may still have dissolved impurities and germs (1); it should be boiled (or purified) before drinking (1).	1+1+1+2
20	Milk is left to stand or boiled and cooled (1); Cream forms a layer on top (1); The cream is removed and churned (1); Butter forms (1); Buttermilk is left behind (1).	1+1+1+1+1
	OR: Sugar dissolves and sand does not (1); Filtration then removes sand (1); Evaporation recovers sugar (1); Handpicking tiny grains is slow (1); Choose the method by property (1).	1+1+1+1+1

Mistakes to avoid:

- Choosing filtration for salt in water; dissolved salt passes through the filter, so **evaporation** is needed.
- Mixing **sedimentation** (settling) with **decantation** (pouring off the clear liquid).
- Mixing **threshing** (beating stalks to free grain) with winnowing (blowing away husk).
- Calling the liquid that passes through filter paper the residue; it is the **filtrate**.
- Using a magnet to separate sand or salt; only magnetic substances like iron are attracted.
- Believing that filtered or settled water is always safe to drink.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** spread and heat it in sun	Drying.
2	**(a)** deposit crystals	Excess solute crystallises.
3	**(b)** iron filings, salt and sand	Magnet removes iron, water dissolves salt.
4	**(a)** evaporation	Gets salt.
5	**(a)** salt passes through filter paper	Dissolved.
6	**(a)** separating iron by magnet	Only magnetism.
7	**(a)** filtration	No change.
8	**(c)** clear filtrate	Clean separation.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The lighter fat collects as butter, so R explains A.
10	**(c)** A is true but R is false.	A is true. R is false: sand is non-magnetic.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Particles pass through (1); Fine solids reach the filtrate (1).	1+1
12	Solubility decreases (1); Excess solute separates out (1).	1+1
13	Mud particles are insoluble and stay on the paper (1); Salt is dissolved and passes through with the water (1).	1+1
14	They are dissolved (1); They are smaller than the pores (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cream lighter (1); Used for butter (1); Skimmed milk remains (1).	1+1+1
16	Moisture adds mass (1); Pure solid obtained (1); Accurate result (1).	1+1+1
17	Filtration (1); Fluff stays (1); Water passes (1).	1+1+1
	OR: Filter to get chalk (1); Evaporate filtrate for sugar (1); Dry chalk (1).	1+1+1
18	Seeds crushed (1); Oil pressed out (1); Filtered (1).	1+1+1
	OR: Remove unwanted stones (1); Iron pieces (1); Quality and safety (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Evaporation (1). (ii) Condensation (1). (iii) Salt (1). (iv) Heat the salt solution; salt remains in the dish after the water evaporates (1) and the vapour is cooled so that it condenses into pure water (1).	1+1+1+2
20	Filtration works for sand-water (1); It fails for salt-water (1); Because salt dissolves (1); Evaporation works (1); Choose by property (1).	1+1+1+1+1
	OR: Evaporate equal amounts (1); Distilled leaves none (1); Tap leaves a little (1); Pond leaves the most (1); Residue shows impurities (1).	1+1+1+1+1

Mistakes to avoid:

- Choosing filtration for salt in water; dissolved salt passes through the filter, so **evaporation** is needed.
- Mixing **sedimentation** (settling) with **decantation** (pouring off the clear liquid).
- Mixing **threshing** (beating stalks to free grain) with winnowing (blowing away husk).
- Calling the liquid that passes through filter paper the residue; it is the **filtrate**.
- Using a magnet to separate sand or salt; only magnetic substances like iron are attracted.
- Believing that filtered or settled water is always safe to drink.

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